

# AMARDEEP SAHA

Backend Software Engineer

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LinkedIn | GitHub | Portfolio | LeetCode

## Summary

Backend Software Engineer with 3+ years of experience designing and scaling distributed backend systems using Golang, Kafka, PostgreSQL, and Redis. Built a concurrent caching layer that cut database calls by over 90% and Kafka-driven pipelines for real-time data validation and enrichment. Strong foundation in data structures, algorithms, and system design, with a track record of improving latency, throughput, and conversion metrics in production.

## Skills

**Programming Languages:** Golang, Python, C++, JavaScript, TypeScript, SQL, PHP, C, Bash, Solidity

**Backend & Frameworks:** Node.js, Express.js, REST APIs, Apache Kafka, Gin, Redux, Next.js, React

**Databases & Caching:** PostgreSQL, MongoDB, Redis, SQL, Change Data Capture (CDC)

**Tools & Platforms:** Docker, Kubernetes, Git, Devtron, Ethereum, Remix IDE

**Core Competencies:** Distributed Systems, System Design, Microservices, Caching Strategies, Data Structures & Algorithms, Performance Optimization

## Experience

### Indiamart

Noida, India

Backend Software Engineer / June 2023 – Present

- Designed and implemented a WhatsApp-based enquiry generation pipeline that identified user intent and triggered automated engagement, generating INR 1,00,000+ in additional profit and scaling onboarding by 200+ users per day.
- Architected a high-throughput concurrent caching layer in Golang using in-memory storage and Redis, reducing database calls by over 90% and achieving approximately 1ms p95 latency.
- Designed and developed a Kafka-driven location validation and enrichment pipeline in Golang with PostgreSQL, detecting location mismatches exceeding 250 km and triggering asynchronous WhatsApp-based remediation workflows to improve data accuracy and automation.
- Built a distributed Intent-to-Lead pipeline processing 2.15M+ buyer clickstream events/day, using Cassandra for high-throughput activity logging, scheduled ETL for extracting buyer search interests and enriching product/seller metadata, and Kafka-based asynchronous processing for downstream lead creation; converted 14
- Led backend performance optimization across 5+ service design reviews, replacing sequential workflows with asynchronous processing and reducing page load time by 25%.
- Owned automated data enrichment workflows for mobile user enquiries, increasing valid form fill rates by 10% through intelligent extraction techniques.

### QuantumLeap

Remote, India

Research and Development Intern / March 2022 – August 2022

- Engineered a full-stack decentralized application (DAO) supporting 10,000+ members and transactions.
- Optimized gas-efficient smart contracts in Solidity for token distribution and stake allocation logic.
- Implemented frontend-to-blockchain connectivity using Ethers.js and MetaMask.
- Deployed and tested smart contracts on the Ethereum blockchain using Remix IDE.

## Projects

**Video Conferencing Application** | React.js, WebRTC, Socket.IO, Node.js, Redis, SQL, Docker

- Engineered a full-stack, real-time video conferencing application with peer-to-peer audio/video streaming via WebRTC and real-time signaling and chat via Socket.IO.
- Built a secure backend with JWT-based authentication and a room management system, paired with a responsive Tailwind CSS UI.

**Voice-Enabled Task Tracker** | Next.js, TypeScript, Node.js, Express, SQLite, LLMs, Docker

- Built a backend-driven task management system with REST APIs supporting CRUD operations, filtering, and search using Node.js, Express, and SQLite.
- Designed an LLM-powered natural language processing pipeline (Groq SDK) to convert voice input into structured task data, including title, priority, due date, and status.
- Integrated speech-to-text ingestion with backend services, enabling real-time voice-based task creation and Kanban workflows.

**Research Publication: “A Survey Paper on Developments in Blockchain”**

- Authored and published a research paper at IIIT Ranchi, analyzing over 25 papers on the evolution of blockchain consensus mechanisms.

## Education

Indian Institute of Information Technology, Ranchi

Ranchi, India

B.Tech, Computer Science and Engineering | July 2019 – May 2023 | CGPA: 8.45 / 10